

LIFEWS AgriShine(TM) GrowMeal(TM)

SSS 1-3 ASSESSMENT & QUIZ SYSTEM

TEACHER MARKING & MODERATION GUIDE

Purpose: assess knowledge, practical competence, investigation/data skills, and mission-based problem solving without reducing garden learning to recall alone.

Domain	Weight	Primary evidence
Knowledge	15%	Oral/written response, concept explanation
Practical competence	20%	Demonstration, technique, safety, record quality
Investigation/Data	30%	Measurement, table/graph, calculation, interpretation
Mission problem solving	35%	Diagnosis, design, recommendation, evidence-based decision

Performance rubric for open/practical items

Score	Descriptor
3 - Proficient	Method is appropriate, evidence is accurate/traceable, and explanation or decision is supported.
2 - Developing	Method is mostly appropriate with minor errors or incomplete reasoning.
1 - Beginning	Major support is needed; evidence or reasoning is incomplete.
0 - Not demonstrated	No valid evidence or the task is not completed.

Moderation rule: award credit for scientifically or practically valid alternatives when they satisfy the task. Do not award marks for unsupported claims. For Nursery, use oral responses and observed performance where writing is not developmentally appropriate.

ANSWER & MARKING GUIDE 01 - Advanced Production Garden

AQ-401 - Knowledge

Frame a research or engineering question involving yield components, harvest index and resource-use efficiency.

Expected answer / marking basis: Precise question with system boundary and measurable performance outcome.

AQ-402 - Knowledge

Identify one assumption and one constraint relevant to the advanced production plot.

Expected answer / marking basis: Valid assumption and operational/design constraint.

AQ-403 - Knowledge

Explain why validation is distinct from calibration or fitting.

Expected answer / marking basis: Validation tests performance on independent/held-out evidence; calibration/fitting sets model/instrument relation.

AQ-404 - Practical

Execute or demonstrate a technically appropriate measurement of efficiency metric, with instrument ID, unit and QC note.

Expected answer / marking basis: 0-3 technical/QC rubric.

AQ-405 - Practical

Document one calibration, verification or reference check.

Expected answer / marking basis: 0-3 documentation rubric.

AQ-406 - Investigation/Data

Analyze a dataset using an appropriate quantitative metric and state uncertainty.

Expected answer / marking basis: Correct metric, units and uncertainty statement.

AQ-407 - Investigation/Data

Compare at least two scenarios/designs and quantify the performance trade-off.

Expected answer / marking basis: Quantified, defensible comparison.

AQ-408 - Mission

Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

Expected answer / marking basis: Valid sensitivity/stress-test reasoning.

AQ-409 - Mission

Recommend an optimization and define a measurable acceptance criterion.

Expected answer / marking basis: Actionable optimization + measurable criterion.

AQ-410 - Mission

Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Expected answer / marking basis: Technically defensible multi-criteria decision.

ANSWER & MARKING GUIDE 02 - Crop Trial Garden

AQ-411 - Knowledge

Frame a research or engineering question involving replicated treatments, blocks, variability and treatment effects.

Expected answer / marking basis: Precise question with system boundary and measurable performance outcome.

AQ-412 - Knowledge

Identify one assumption and one constraint relevant to the crop trial.

Expected answer / marking basis: Valid assumption and operational/design constraint.

AQ-413 - Knowledge

Explain why validation is distinct from calibration or fitting.

Expected answer / marking basis: Validation tests performance on independent/held-out evidence; calibration/fitting sets model/instrument relation.

AQ-414 - Practical

Execute or demonstrate a technically appropriate measurement of treatment effect, with instrument ID, unit and QC note.

Expected answer / marking basis: 0-3 technical/QC rubric.

AQ-415 - Practical

Document one calibration, verification or reference check.

Expected answer / marking basis: 0-3 documentation rubric.

AQ-416 - Investigation/Data

Analyze a dataset using an appropriate quantitative metric and state uncertainty.

Expected answer / marking basis: Correct metric, units and uncertainty statement.

AQ-417 - Investigation/Data

Compare at least two scenarios/designs and quantify the performance trade-off.

Expected answer / marking basis: Quantified, defensible comparison.

AQ-418 - Mission

Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

Expected answer / marking basis: Valid sensitivity/stress-test reasoning.

AQ-419 - Mission

Recommend an optimization and define a measurable acceptance criterion.

Expected answer / marking basis: Actionable optimization + measurable criterion.

AQ-420 - Mission

Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Expected answer / marking basis: Technically defensible multi-criteria decision.

ANSWER & MARKING GUIDE 03 - Controlled Experiment Garden

AQ-421 - Knowledge

Frame a research or engineering question involving causal variables, factorial effects, confounding and sensitivity.

Expected answer / marking basis: Precise question with system boundary and measurable performance outcome.

AQ-422 - Knowledge

Identify one assumption and one constraint relevant to the controlled experiment.

Expected answer / marking basis: Valid assumption and operational/design constraint.

AQ-423 - Knowledge

Explain why validation is distinct from calibration or fitting.

Expected answer / marking basis: Validation tests performance on independent/held-out evidence; calibration/fitting sets model/instrument relation.

AQ-424 - Practical

Execute or demonstrate a technically appropriate measurement of experimental metric, with instrument ID, unit and QC note.

Expected answer / marking basis: 0-3 technical/QC rubric.

AQ-425 - Practical

Document one calibration, verification or reference check.

Expected answer / marking basis: 0-3 documentation rubric.

AQ-426 - Investigation/Data

Analyze a dataset using an appropriate quantitative metric and state uncertainty.

Expected answer / marking basis: Correct metric, units and uncertainty statement.

AQ-427 - Investigation/Data

Compare at least two scenarios/designs and quantify the performance trade-off.

Expected answer / marking basis: Quantified, defensible comparison.

AQ-428 - Mission

Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

Expected answer / marking basis: Valid sensitivity/stress-test reasoning.

AQ-429 - Mission

Recommend an optimization and define a measurable acceptance criterion.

Expected answer / marking basis: Actionable optimization + measurable criterion.

AQ-430 - Mission

Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Expected answer / marking basis: Technically defensible multi-criteria decision.

ANSWER & MARKING GUIDE 04 - Irrigation Engineering Garden

AQ-431 - Knowledge

Frame a research or engineering question involving flow, head, pipe loss, uniformity and pumping efficiency.

Expected answer / marking basis: Precise question with system boundary and measurable performance outcome.

AQ-432 - Knowledge

Identify one assumption and one constraint relevant to the irrigation system.

Expected answer / marking basis: Valid assumption and operational/design constraint.

AQ-433 - Knowledge

Explain why validation is distinct from calibration or fitting.

Expected answer / marking basis: Validation tests performance on independent/held-out evidence; calibration/fitting sets model/instrument relation.

AQ-434 - Practical

Execute or demonstrate a technically appropriate measurement of engineering metric, with instrument ID, unit and QC note.

Expected answer / marking basis: 0-3 technical/QC rubric.

AQ-435 - Practical

Document one calibration, verification or reference check.

Expected answer / marking basis: 0-3 documentation rubric.

AQ-436 - Investigation/Data

Analyze a dataset using an appropriate quantitative metric and state uncertainty.

Expected answer / marking basis: Correct metric, units and uncertainty statement.

AQ-437 - Investigation/Data

Compare at least two scenarios/designs and quantify the performance trade-off.

Expected answer / marking basis: Quantified, defensible comparison.

AQ-438 - Mission

Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

Expected answer / marking basis: Valid sensitivity/stress-test reasoning.

AQ-439 - Mission

Recommend an optimization and define a measurable acceptance criterion.

Expected answer / marking basis: Actionable optimization + measurable criterion.

AQ-440 - Mission

Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Expected answer / marking basis: Technically defensible multi-criteria decision.

ANSWER & MARKING GUIDE 05 - Water Productivity Garden

AQ-441 - Knowledge

Frame a research or engineering question involving water balance, ET, IWP and ET water productivity.

Expected answer / marking basis: Precise question with system boundary and measurable performance outcome.

AQ-442 - Knowledge

Identify one assumption and one constraint relevant to the water-productivity dataset.

Expected answer / marking basis: Valid assumption and operational/design constraint.

AQ-443 - Knowledge

Explain why validation is distinct from calibration or fitting.

Expected answer / marking basis: Validation tests performance on independent/held-out evidence; calibration/fitting sets model/instrument relation.

AQ-444 - Practical

Execute or demonstrate a technically appropriate measurement of water-productivity metric, with instrument ID, unit and QC note.

Expected answer / marking basis: 0-3 technical/QC rubric.

AQ-445 - Practical

Document one calibration, verification or reference check.

Expected answer / marking basis: 0-3 documentation rubric.

AQ-446 - Investigation/Data

Analyze a dataset using an appropriate quantitative metric and state uncertainty.

Expected answer / marking basis: Correct metric, units and uncertainty statement.

AQ-447 - Investigation/Data

Compare at least two scenarios/designs and quantify the performance trade-off.

Expected answer / marking basis: Quantified, defensible comparison.

AQ-448 - Mission

Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

Expected answer / marking basis: Valid sensitivity/stress-test reasoning.

AQ-449 - Mission

Recommend an optimization and define a measurable acceptance criterion.

Expected answer / marking basis: Actionable optimization + measurable criterion.

AQ-450 - Mission

Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Expected answer / marking basis: Technically defensible multi-criteria decision.

ANSWER & MARKING GUIDE 06 - Advanced Soilless Garden

AQ-451 - Knowledge

Frame a research or engineering question involving pH/EC dynamics, flow and residence time.

Expected answer / marking basis: Precise question with system boundary and measurable performance outcome.

AQ-452 - Knowledge

Identify one assumption and one constraint relevant to the advanced soilless unit.

Expected answer / marking basis: Valid assumption and operational/design constraint.

AQ-453 - Knowledge

Explain why validation is distinct from calibration or fitting.

Expected answer / marking basis: Validation tests performance on independent/held-out evidence; calibration/fitting sets model/instrument relation.

AQ-454 - Practical

Execute or demonstrate a technically appropriate measurement of soilless metric, with instrument ID, unit and QC note.

Expected answer / marking basis: 0-3 technical/QC rubric.

AQ-455 - Practical

Document one calibration, verification or reference check.

Expected answer / marking basis: 0-3 documentation rubric.

AQ-456 - Investigation/Data

Analyze a dataset using an appropriate quantitative metric and state uncertainty.

Expected answer / marking basis: Correct metric, units and uncertainty statement.

AQ-457 - Investigation/Data

Compare at least two scenarios/designs and quantify the performance trade-off.

Expected answer / marking basis: Quantified, defensible comparison.

AQ-458 - Mission

Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

Expected answer / marking basis: Valid sensitivity/stress-test reasoning.

AQ-459 - Mission

Recommend an optimization and define a measurable acceptance criterion.

Expected answer / marking basis: Actionable optimization + measurable criterion.

AQ-460 - Mission

Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Expected answer / marking basis: Technically defensible multi-criteria decision.

ANSWER & MARKING GUIDE 07 - Sensor & Monitoring Garden

AQ-461 - Knowledge

Frame a research or engineering question involving accuracy, precision, calibration, MAE/RMSE and network design.

Expected answer / marking basis: Precise question with system boundary and measurable performance outcome.

AQ-462 - Knowledge

Identify one assumption and one constraint relevant to the monitoring station.

Expected answer / marking basis: Valid assumption and operational/design constraint.

AQ-463 - Knowledge

Explain why validation is distinct from calibration or fitting.

Expected answer / marking basis: Validation tests performance on independent/held-out evidence; calibration/fitting sets model/instrument relation.

AQ-464 - Practical

Execute or demonstrate a technically appropriate measurement of monitoring metric, with instrument ID, unit and QC note.

Expected answer / marking basis: 0-3 technical/QC rubric.

AQ-465 - Practical

Document one calibration, verification or reference check.

Expected answer / marking basis: 0-3 documentation rubric.

AQ-466 - Investigation/Data

Analyze a dataset using an appropriate quantitative metric and state uncertainty.

Expected answer / marking basis: Correct metric, units and uncertainty statement.

AQ-467 - Investigation/Data

Compare at least two scenarios/designs and quantify the performance trade-off.

Expected answer / marking basis: Quantified, defensible comparison.

AQ-468 - Mission

Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

Expected answer / marking basis: Valid sensitivity/stress-test reasoning.

AQ-469 - Mission

Recommend an optimization and define a measurable acceptance criterion.

Expected answer / marking basis: Actionable optimization + measurable criterion.

AQ-470 - Mission

Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Expected answer / marking basis: Technically defensible multi-criteria decision.

ANSWER & MARKING GUIDE 08 - Smart Irrigation Demonstrator

AQ-471 - Knowledge

Frame a research or engineering question involving thresholds, hysteresis, feedback and adaptive control.

Expected answer / marking basis: Precise question with system boundary and measurable performance outcome.

AQ-472 - Knowledge

Identify one assumption and one constraint relevant to the smart irrigation controller.

Expected answer / marking basis: Valid assumption and operational/design constraint.

AQ-473 - Knowledge

Explain why validation is distinct from calibration or fitting.

Expected answer / marking basis: Validation tests performance on independent/held-out evidence; calibration/fitting sets model/instrument relation.

AQ-474 - Practical

Execute or demonstrate a technically appropriate measurement of control metric, with instrument ID, unit and QC note.

Expected answer / marking basis: 0-3 technical/QC rubric.

AQ-475 - Practical

Document one calibration, verification or reference check.

Expected answer / marking basis: 0-3 documentation rubric.

AQ-476 - Investigation/Data

Analyze a dataset using an appropriate quantitative metric and state uncertainty.

Expected answer / marking basis: Correct metric, units and uncertainty statement.

AQ-477 - Investigation/Data

Compare at least two scenarios/designs and quantify the performance trade-off.

Expected answer / marking basis: Quantified, defensible comparison.

AQ-478 - Mission

Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

Expected answer / marking basis: Valid sensitivity/stress-test reasoning.

AQ-479 - Mission

Recommend an optimization and define a measurable acceptance criterion.

Expected answer / marking basis: Actionable optimization + measurable criterion.

AQ-480 - Mission

Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Expected answer / marking basis: Technically defensible multi-criteria decision.

ANSWER & MARKING GUIDE 09 - Research & Design Garden

AQ-481 - Knowledge

Frame a research or engineering question involving problem definition, requirements, concept selection and prototype testing.

Expected answer / marking basis: Precise question with system boundary and measurable performance outcome.

AQ-482 - Knowledge

Identify one assumption and one constraint relevant to the design brief.

Expected answer / marking basis: Valid assumption and operational/design constraint.

AQ-483 - Knowledge

Explain why validation is distinct from calibration or fitting.

Expected answer / marking basis: Validation tests performance on independent/held-out evidence; calibration/fitting sets model/instrument relation.

AQ-484 - Practical

Execute or demonstrate a technically appropriate measurement of design metric, with instrument ID, unit and QC note.

Expected answer / marking basis: 0-3 technical/QC rubric.

AQ-485 - Practical

Document one calibration, verification or reference check.

Expected answer / marking basis: 0-3 documentation rubric.

AQ-486 - Investigation/Data

Analyze a dataset using an appropriate quantitative metric and state uncertainty.

Expected answer / marking basis: Correct metric, units and uncertainty statement.

AQ-487 - Investigation/Data

Compare at least two scenarios/designs and quantify the performance trade-off.

Expected answer / marking basis: Quantified, defensible comparison.

AQ-488 - Mission

Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

Expected answer / marking basis: Valid sensitivity/stress-test reasoning.

AQ-489 - Mission

Recommend an optimization and define a measurable acceptance criterion.

Expected answer / marking basis: Actionable optimization + measurable criterion.

AQ-490 - Mission

Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Expected answer / marking basis: Technically defensible multi-criteria decision.

ANSWER & MARKING GUIDE 10 - Integrated Food Systems Laboratory

AQ-491 - Knowledge

Frame a research or engineering question involving system boundaries, balances, time-varying mismatch, cascades and scenarios.

Expected answer / marking basis: Precise question with system boundary and measurable performance outcome.

AQ-492 - Knowledge

Identify one assumption and one constraint relevant to the integrated system model.

Expected answer / marking basis: Valid assumption and operational/design constraint.

AQ-493 - Knowledge

Explain why validation is distinct from calibration or fitting.

Expected answer / marking basis: Validation tests performance on independent/held-out evidence; calibration/fitting sets model/instrument relation.

AQ-494 - Practical

Execute or demonstrate a technically appropriate measurement of integrated-system metric, with instrument ID, unit and QC note.

Expected answer / marking basis: 0-3 technical/QC rubric.

AQ-495 - Practical

Document one calibration, verification or reference check.

Expected answer / marking basis: 0-3 documentation rubric.

AQ-496 - Investigation/Data

Analyze a dataset using an appropriate quantitative metric and state uncertainty.

Expected answer / marking basis: Correct metric, units and uncertainty statement.

AQ-497 - Investigation/Data

Compare at least two scenarios/designs and quantify the performance trade-off.

Expected answer / marking basis: Quantified, defensible comparison.

AQ-498 - Mission

Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

Expected answer / marking basis: Valid sensitivity/stress-test reasoning.

AQ-499 - Mission

Recommend an optimization and define a measurable acceptance criterion.

Expected answer / marking basis: Actionable optimization + measurable criterion.

AQ-500 - Mission

Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Expected answer / marking basis: Technically defensible multi-criteria decision.

LEVEL MODERATION & REPORTING

Recommended reporting: maintain the four domain scores separately as well as a weighted total. A student may have a strong overall total but still need targeted support in practical technique, data interpretation or problem solving.

Suggested achievement bands: 85-100 Advanced; 70-84 Secure; 55-69 Developing; 40-54 Emerging; below 40 Requires targeted support. Schools may adjust bands to their approved assessment policy.

Teacher moderation notes: